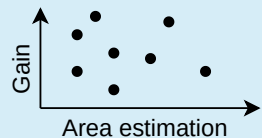


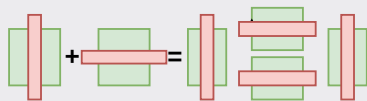
Inputs

Electrical design space
(from pre-layout exploration)



+

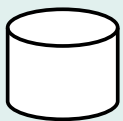
PCell library
(ShapelC-Cellkit)



- Parametrized layouts
- DRC/LVS Clean

+

Physical LUT
(from PCells)



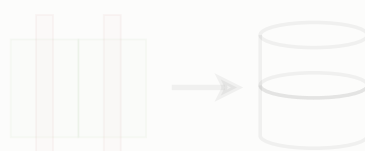
- Parasitics vs geometry
- Pre-characterized
- Reusable

Layout-aware exploration (combine electrical + physical)

For each candidate
(from electrical design space)



Query physical LUTs
(based on layout parameters)



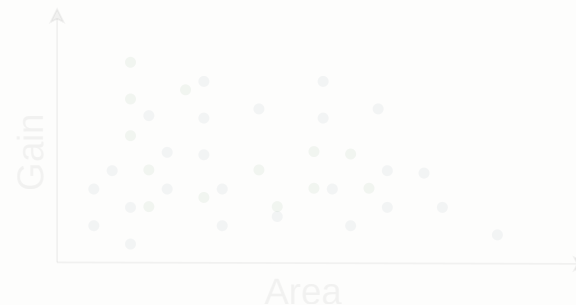
Compose layout-aware model
(electrical + parasitics)

MNA

Evaluate (AC, etc)

Specs filter

Results (Layout-aware design space)



Selected design



- ✓ Meets specs
- ✓ DRC/LVS clean
- ✓ Accurate parasitics

Key benefits

- Realistic performance (include parasitics)
- Layout-dependent trade-offs
- Reuse physical LUTs
- Fast exploration (no full PEX in the loop)